

Qatar-10b

R-0031

Benchmark run · workflow W8-benchmark-deep · record deep-bench_Qatar-10_260426 · created 2026-07-08T18:07:52+00:00

BENCHMARK: INACCURATE

Results

Mid-Transit Time (Tmid)	2461156.858 +/- 0.014 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.17 +/- 0.05
Transit depth (Rp/Rs)^2	2.88 +/- 1.68 [%]
Orbital Inclination (inc)	84.62 +/- 2.61
Airmass coefficient 1 (a1)	0.69 +/- 0.44
Airmass coefficient 2 (a2)	0.14 +/- 0.25
Scatter in the residuals of the lightcurve fit is	57.15 %
Best Comparison Star	None
Optimal Aperture	3.0
Optimal Annulus	3.0
Transit Duration (day)	0.048 +/- 0.016

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.17 ± 0.05 vs 0.1265 (deviation 0.87σ)
Duration fit vs published	0.048 d vs 0.1154 d (deviation 4.21σ)
Mid-time O–C	19.2 min
Depth SNR	3.4
Residual scatter	57.15 %

Light curve

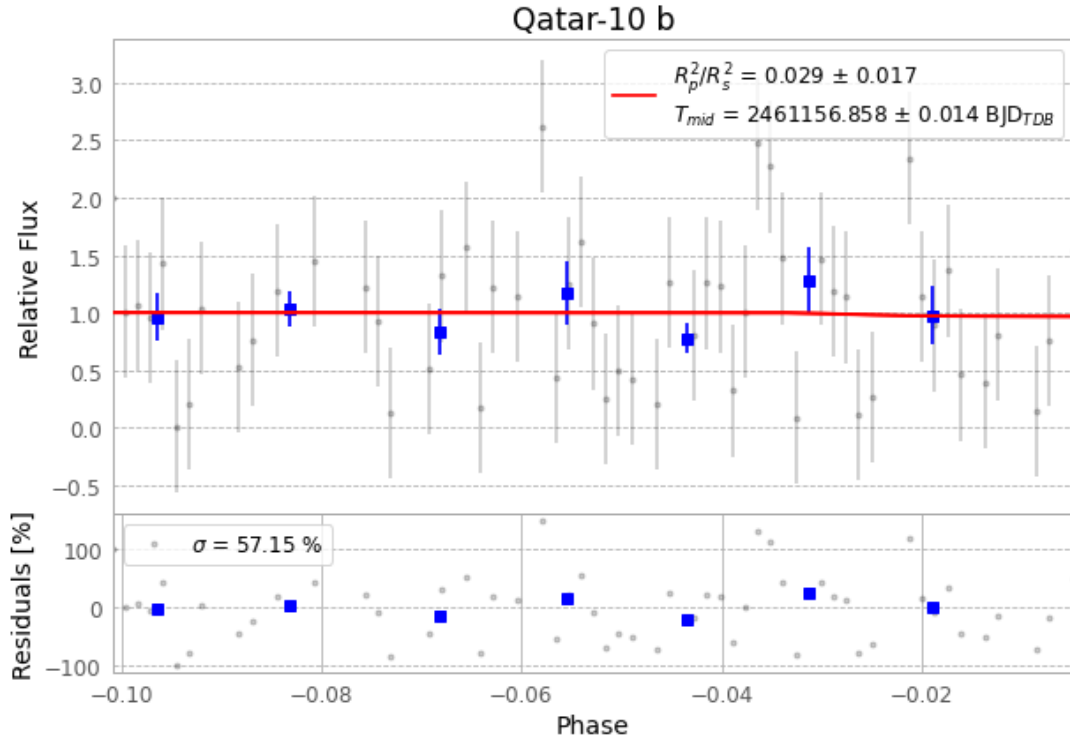


Figure 1 — FinalLightCurve_Qatar-10 b_2026-04-25.png (SHA-256 1c3598f0d94bff36...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [309, 289], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 7a5752f7d923bc2d...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13

Data provenance

78 FITS frames (51 MB), Qatar-10260426043315.FITS → Qatar-10260426082415.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-Qatar-10-260426.log (ab161d18bd08...), FinalLightCurve_Qatar-10 b_2026-04-25.png (1c3598f0d94b...), FinalLightCurve_Qatar-10 b_2026-04-25.csv (4568a943c862...)

Compute resources

Wall time 173.6 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-15 16:40Z.